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Coil device

Abstract

A coil device capable of reducing magnetic coupling between conductors while ensuring good inductance characteristics is provided. A coil device **10** includes a first core **20**, a second core **30** combined with the first core **20**, and a first conductor **40** and a second conductor **50** arranged adjacent to each other between the first core **20** and the second core **30**. At least one of the first core **20** and the second core **30** includes a center leg portion **23** (or a center leg portion **33**) and a pair of outer leg portions **22***a*, **22***b* (or outer leg portions **32***a*, **32***b*) arranged on both sides of the center leg portion **23** (or center leg portion **33**), and a magnetic body is arranged between the first conductor **40** and the second conductor **50.**

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Background/Summary

BACKGROUND OF THE INVENTION
Field of the Invention
(1) The present invention relates to a coil device used as a coupled inductor or the like.
Description of the Related Art
(2) A coil device called a coupled inductor may be used as a smoothing coil for a switching power supply such as a DC/DC converter. The coupled inductor includes a pair of conductors and the conductors are magnetically coupled with each other by a predetermined coupling coefficient. In recent years, there has been a demand for a coupled inductor having a relatively small coupling

coefficient, and examples of a technique for implementing this type of coupled inductor include a technique described in JP-A-2009-16797 (Patent Literature 1).

(3) A coil device described in Patent Literature 1 includes a first core, a second core combined with the first core, and a pair of conductors arranged between the first core and the second core. Each of the first core and the second core includes a center leg portion and a pair of outer leg portions arranged on both sides of the center leg portion. By increasing a gap amount between the first core and the second core at positions of the outer leg portions, it is possible to reduce the coupling coefficient between the conductors.

(4) However, when the gap amount between the first core and the second core is increased as in the coil device described in Patent Literature 1, an inductance value becomes low and good inductance characteristics cannot be obtained.

SUMMARY OF THE INVENTION

(5) The present invention is made in view of such a circumstance and an object of the present invention is to provide a coil device capable of reducing magnetic coupling between conductors while ensuring good inductance characteristics.

(6) In order to achieve the above-mentioned object, the coil device according to the present invention includes: a first core; a second core combined with the first core; and a first conductor and a second conductor arranged adjacent to each other between the first core and the second core, in which at least one of the first core and the second core includes a center leg portion and a pair of outer leg portions arranged on both sides of the center leg portion, and a magnetic body is arranged between the first conductor and the second conductor.

(7) In the coil device according to the present invention, the magnetic body is arranged between the first conductor and the second conductor. In this case, coupling between the first conductor and the second conductor is lower than that in a case where no magnetic body is arranged between the first conductor and the second conductor, and it is possible to reduce magnetic coupling between the first conductor and the second conductor. By arranging the magnetic body between the first conductor and the second conductor, the magnetic body contributes to inductance of the coil device, and the inductance value of the whole coil device can be increased. Therefore, according to the coil device according to the present invention, it is possible to reduce the magnetic coupling between the conductors while ensuring good inductance characteristics.

(8) A ratio of a cross-sectional area of the center leg portion to a cross-sectional area of the outer leg portions may be 1:1 to 1:4. In this case, the center leg portion functions as the above-mentioned magnetic body arranged between the first conductor and the second conductor. By such a configuration, it is possible to sufficiently reduce a coupling coefficient between the first conductor and the second conductor, and reduce the magnetic coupling between the conductors while ensuring good inductance characteristics.

(9) A ratio of a width of the center leg portion to a width of the outer leg portions may be 1:1 to 1:4. When the ratio of the cross-sectional area of the center leg portion to the cross-sectional area of the outer leg portions is 1:1 to 1:4, by setting the ratio of the width of the center leg portion to the width of the outer leg portions within the above-mentioned range, it is possible to match protruding widths of the center leg portion and the outer leg portions with each other, and a symmetry between the first core and the second core is good. Therefore, the coil device having good inductance characteristics can be effectively obtained.

(10) The first core may be arranged above the second core and be larger than the second core. By such a configuration, when the first conductor and the second conductor are arranged between the first core and the second core, it is possible to prevent the first conductor and the second conductor from protruding outside the first core, which can contribute to miniaturization of the coil device.

(11) The first core may include a first center leg portion and a first outer leg portion, and the second core may include a second center leg portion and a second outer leg portion. A first recess portion may be formed between the first center leg portion and the first outer leg portion, and a second

recess portion may be formed between the second center leg portion and the second outer leg portion. A height of the first center leg portion from a bottom surface of the first recess portion may different from a height of the second center leg portion from a bottom surface of the second recess portion. In this case, when the first core and the second core are combined, a joint portion between the first center leg portion and the second center leg portion is arranged at any height position between the bottom surface of the first recess portion and the bottom surface of the second recess portion, so that the first center leg portion and the second center leg portion can be arranged between the first conductor and the second conductor. Therefore, in this case as well, it is possible to reduce the magnetic coupling between the first conductor and the second conductor while ensuring good inductance characteristics.

(12) A first mounting portion may be provided at both end portions in a longitudinal direction of the first conductor, and a second mounting portion may be provided at both end portions in a longitudinal direction of the second conductor. The first mounting portion may extend toward a side on which one of the pair of outer leg portions is arranged, and the second mounting portion may extend toward a side on which the other one of the pair of outer leg portions is arranged, which is opposite to that of the first mounting portion. By such a configuration, the first mounting portion and the second mounting portion can be separated from each other, and it is possible to prevent a short circuit failure from occurring between the first mounting portion and the second mounting portion. It is possible to ensure a sufficient mounting area for each of the first mounting portion and the second mounting portion, and the coil device can be firmly fixed to a mounting substrate.

(13) At least one of the first core and the second core may contain a metallic magnetic material. By such a configuration, the coupling coefficient between the first conductor and the second conductor can be effectively reduced to a desired value.

(14) The first conductor and the second conductor may be made of conductive plate sheets. By such a configuration, permissible currents flowing through the first conductor and the second conductor can be increased.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. **1**A is a perspective view of a coil device according to a first embodiment of the present invention.

(2) FIG. **1**B is a side view of the coil device shown in FIG. **1**A when viewed from an X-axis direction.

(3) FIG. **1**C is a side view of the coil device shown in FIG. **1**A when viewed from a Y-axis direction.

(4) FIG. **2** is an exploded perspective view of the coil device shown in FIG. **1**A.

(5) FIG. **3**A is a cross-sectional view taken along a line IIIA-IIIA of the coil device shown in FIG. **1**A.

(6) FIG. **3**B is a cross-sectional view taken along a line IIIB-IIIB of the coil device shown in FIG. **1**A.

(7) FIG. **3**C is a cross-sectional view taken along a line IIIC-IIIC of the coil device shown in FIG. **3**A.

(8) FIG. **4** is a side view of a coil device according to a second embodiment of the present invention when viewed from an X-axis direction.

(9) FIG. **5** is a perspective view of a coil device according to a third embodiment of the present invention.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

(10) Hereinafter, the present invention will be described based on embodiments shown in the drawings.
First Embodiment
(11) As shown in FIG. **1**A, a coil device **10** according to a first embodiment of the present invention includes a first core **20**, a second core **30** combined with the first core **20**, and a first conductor **40** and a second conductor **50** arranged adjacent to each other between the first core **20** and the second core **30**. The coil device **10** is, for example, a coupled inductor and is used as a smoothing coil for a switching power supply such as a DC/DC converter. The above-mentioned switching power supply is used in a power supply circuit in a server, a mobile terminal, or the like.
(12) The coil device **10** has a substantially rectangular parallelepiped shape as a whole. A correlation among a width W**1** in an X-axis direction of the coil device **10** (corresponding to a width in the X-axis direction of the first core **20**), a width W**2** in a Y-axis direction of the coil device **10**, and a height H**1** of the coil device **10** is W**2**>W**1**>H**1**. That is, an overall shape of the coil device **10** is a substantially flat shape (thin shape). The width W**1** in the X-axis direction is preferably 5.0 to 20.0 mm. The width W**2** in the Y-axis direction is preferably 5.0 to 20.0 mm. The height H**1** is preferably 2.0 to 10.0 mm.
(13) The first core **20** and the second core **30** contain a metallic magnetic material and are obtained by, for example, compression-molding metallic magnetic powder containing metallic magnetic particles. Although the metallic magnetic material is not particularly limited, examples thereof include Fe—Ni alloy powder, Fe—Si alloy powder, Fe—Si—Cr alloy powder, Fe—Co alloy powder, Fe—Si—Al alloy powder, amorphous iron, etc. However, materials constituting the first core **20** and the second core **30** are not limited to these, and the first core **20** and the second core **30** may be made of, for example, ferrite. Examples of ferrite include Ni—Zn-based ferrite, Mn—Zn-based ferrite, etc. Relative permeabilities of the first core **20** and the second core **30** are preferably 40 to 60. A material constituting the first core **20** and a material constituting the second core **30** may be the same with or different from each other.
(14) As shown in FIG. **2**, the first core **20** and the second core **30** have shapes corresponding to each other, and the second core **30** is arranged below the first core **20** in a Z-axis direction. The first core **20** and the second core **30** both have an E-shaped cross section when viewed in a Y-Z cross section, and constitute a so-called E-shaped core.
(15) The first core **20** includes a first base portion **21**, a pair of first outer leg portions **22***a*, **22***b*, a first center leg portion **23**, and a pair of first recess portions **24***a*, **24***b*. The first base portion **21** has a substantially flat plate shape and is formed in a longitudinal shape in the Y-axis direction.
(16) The pair of first outer leg portions **22***a* and **22***b* are arranged on both sides of the first center leg portion **23**. The pair of first outer leg portions **22***a* and **22***b* have the same shape, and protrude downward in the Z-axis direction from both ends in the Y-axis direction of the first base portion **21**. As shown in FIG. **1**B, a length L**1** (a height of the first outer leg portion **22***a* from a bottom surface of the first recess portion **24***a*, which will be described later) in the Z-axis direction of the first outer leg portion **22***a* arranged on one side in the Y-axis direction is equal to a length L**2** (a height of the first outer leg portion **22***b* from a bottom surface of the first recess portion **24***b*, which will be described later) in the Z-axis direction of the first outer leg portion **22***b* arranged on the other side in the Y-axis direction.
(17) The first center leg portion **23** protrudes downward in the Z-axis direction from a center in the Y-axis direction of the first base portion **21**. A length L**3** (a height of the first center leg portion **23** from the bottom surfaces of the recess portions **24***a* and **24***b*, which will be described later) in the Z-axis direction of the first center leg portion **23** is equal to the length L**1** and L**2** in the Z-axis direction of the first outer leg portions **22***a* and **22***b*.
(18) As shown in FIG. **2**, the second core **30** includes a second base portion **31**, a pair of second outer leg portions **32***a*, **32***b*, a second center leg portion **33**, and a pair of second recess portions **34***a*, **34***b*. The second base portion **31** has a substantially flat plate shape and is formed in a

longitudinal shape in the Y-axis direction.

(19) The pair of second outer leg portions **32***a*, **32***b* are arranged on both sides of the second center leg portion **33**. The pair of second outer leg portions **32***a*, **32***b* have the same shape, and protrude upwards in the Z-axis direction from both ends in the Y-axis direction of the second base portion **31**. As shown in FIG. **1**B, a length L**4** (a height of the second outer leg portion **32***a* from a bottom surface of the second recess portion **34***a*, which will be described later) in the Z-axis direction of the second outer leg portion **32***a* arranged on one side in the Y-axis direction is equal to a length L**5** (a height of the second outer leg portion **32***b* from a bottom surface of the second recess portion **34***b*, which will be described later) in the Z-axis direction of the second outer leg portion **32***b* arranged on the other side in the Y-axis direction.

(20) The second center leg portion **33** protrudes upwards in the Z-axis direction from a center in the Y-axis direction of the second base portion **31**. A length L**6** (a height of the second center leg portion **33** from the bottom surfaces of the recess portions **34***a* and **34***b*, which will be described later) in the Z-axis direction of the second center leg portion **33** is equal to the length L**4** and L**5** in the Z-axis direction of the second outer leg portions **32***a*, **32***b*.

(21) As shown in FIG. **3**C, the second center leg portion **33** extends continuously along the X-axis direction from one end to the other end of the second core **30** in the X-axis direction and separates the first conductor **40** and the second conductor **50**. Similarly, each of the pair of outer leg portions **32***a*, **32***b* also extends continuously along the X-axis direction from the one end to the other end of the second core **30** in the X-axis direction. Although detailed illustration is omitted, the same applies to the first center leg portion **23** and the pair of first outer leg portions **22***a* and **22***b*.

(22) As shown in FIGS. **1**A and **1**C, the first core **20** is arranged above the second core **30** and is larger than the second core **30**. More specifically, a length of the first core **20** in the Y-axis direction is substantially equal to a length of the second core **30** in the Y-axis direction, but a length of the first core **20** in the X-axis direction is larger than a length of the second core **30** in the X-axis direction.

(23) In an example shown in FIG. **1**C, one end in the X-axis direction of the first core **20** is located outside one end in the X-axis direction of the second core **30** in the X-axis direction, and the other end in the X-axis direction of the first core **20** is located outside the other end in the X-axis direction of the second core **30** in the X-axis direction. Therefore, the first core **20** protrudes outside the second core **30** in the X-axis direction. When the coil device **10** is viewed from above, the second core **30**, the first conductor **40** (first mounting portions **42***a*, **42***b* in particular), and the second conductor **50** (second mounting portions **52***a*, **52***b* in particular) are hidden (covered) by the first core **20** and cannot be seen.

(24) The second mounting portion **52***a* of the second conductor **50** (or the first mounting portion **42***a* of the first conductor **40** (not shown)) is arranged below the one end portion in the X-axis direction of the first core **20**, and the second mounting portion **52***b* (or the first mounting portion **42***b* of the first conductor **40** (not shown)) of the second conductor **50** is arranged below the other end portion in the X-axis direction of the first core **20**.

(25) With reference to one end in the X-axis direction of the second core **30**, a protrusion length L**7** toward the one end in the X-axis direction of the first core **20** is approximately equal to or greater than a plate thickness T**1** of the second conductor **50** (L**7**≥T**1**). The same applies to a protrusion length toward the other end in the X-axis direction of the first core **20** when the other end in the X-axis direction of the second core **30** is used as a reference.

(26) As shown in FIG. **2**, the first recess portion **24***a* is formed between the first center leg portion **23** and the first outer leg portion **22***a*, and the first recess portion **24***b* is formed between the first center leg portion **23** and the first outer leg portion **22***b*. The first recess portion **24***a* and the first recess portion **24***b* are adjacent to each other in the Y-axis direction with the first center leg portion **23** interposed therebetween. A depth of the first recess portion **24***a* in the Z-axis direction is substantially equal to a depth of the first recess portion **24***b* in the Z-axis direction.

(27) The second recess portion **34***a* is formed between the second center leg portion **33** and the second outer leg portion **32***a*, and the second recess portion **34***b* is formed between the second center leg portion **33** and the second outer leg portion **32***b*. The second recess portion **34***a* and the second recess portion **34***b* are adjacent to each other in the Y-axis direction with the second center leg portion **33** interposed therebetween. A depth of the second recess portion **34***a* in the Z-axis direction is substantially equal to a depth of the second recess portion **34***b* in the Z-axis direction.
(28) As shown in FIG. **1**B, when the first core **20** and the second core **30** are combined in the Z-axis direction, a first gap **61** is formed between the first center leg portion **23** and the second center leg portion **33**, and a second gap **62** is formed between each of the first outer leg portions **22***a*, **22***b* and a corresponding one of the second outer leg portions **32***a*, **32***b*. A width in the Z-axis direction of the first gap **61** is substantially equal to a width in the Z-axis direction of the second gap **62**.
(29) The widths in the Z-axis direction (gap intervals) of the gaps **61**, **62** are sufficiently small with respect to the lengths L**1**, L**4** in the Z-axis direction of the outer leg portions **22***a*, **32***a*, the lengths L**2**, L**5** in the Z-axis direction of the outer leg portions **22***b*, **32***b*, or the lengths L**3**, L**6** in the Z-axis direction of the center leg portions **23**, **33**, and are preferably 0.0 to 0.3 mm. An inductance value of the coil device **10** can be controlled by adjusting the widths in the Z-axis direction of the gaps **61**, **62**.
(30) The first core **20** and the second core **30** are combined by joining the first outer leg portions **22***a*, **22***b* of the first core **20** and the second outer leg portions **32***a*, **32***b* of the second core **30** with a joining material such as an adhesive. For example, by using Micropearl (Sekisui Chemical Co., Ltd.) or a resin containing resin beads as the adhesive, the gaps **61**, **62** can be easily formed between the first core **20** and the second core **30**. The first center leg portion **23** and the second center leg portion **33** may be joined by the above-mentioned adhesive, or only the first outer leg portion **22***a* (or the first outer leg portion **22***b*) and the second outer leg portion **32***a* (or the second outer leg portion **32***b*) may be joined by the above-mentioned adhesive.
(31) A joint portion between the first outer leg portion **22***a* and the second outer leg portion **32***a*, a joint portion between the first outer leg portion **22***b* and the second outer leg portion **32***b*, and a joint portion between the first center leg portion **23** and the second center leg portion **33** are arranged in a region in the Z-axis direction between the bottom surfaces of the recess portions **24***a*, **24***b* and the bottom surfaces of the recess portions **34***a*, **34***b*.
(32) As shown in FIG. **1**A, one ends in the X-axis direction of the first outer leg portions **22***a*, **22***b* are located (protruding) outside one ends in the X-axis direction of the second outer leg portions **32***a*, **32***b* in the X-axis direction, and one end in the X-axis direction of the first center leg portion **23** is located (protruded) outside one end in the X-axis direction of the second center leg portion **33** in the X-axis direction. Although detailed illustration is omitted, the other ends in the X-axis direction of the first outer leg portions **22***a*, **22***b* are located (protruding) outside the other ends in the X-axis direction of the second outer leg portions **32***a*, **32***b* in the X-axis direction, and the other end in the X-axis direction of the first center leg portion **23** is located (protruding) outside the other end in the X-axis direction of the second center leg portion **33** in the X-axis direction.
(33) As shown in FIG. **2**, the first conductor **40** and the second conductor **50** have the same shape and are arranged adjacent to each other with a predetermined distance in the Y-axis direction. An interval between the first conductor **40** and the second conductor **50** is equal to or greater than widths in the Y-axis direction of the center leg portions **23**, **33** (see FIG. **1**B).
(34) The first conductor **40** and the second conductor **50** are made of conductive plate sheets (conductor plates) and have a substantially U shape. Widths in the Y-axis direction of the conductors **40**, **50** are larger than the width in the Y-axis direction of each of the center leg portions **23**, **33** and the outer leg portions **22***a*, **32***a* (or the outer leg portions **22***b*, **32***b*). Examples of materials constituting the conductors **40**, **50** include good conductors of metals such as copper and copper alloys, silver, and nickel, but materials of the conductors are not particularly limited. The conductors **40**, **50** are formed by, for example, machining a metal plate member. However, a

method for forming the conductors **40**, **50** is not limited thereto and may be appropriately changed. As shown in FIG. **3**B, a length in the X-axis direction of the first conductor **40** (the same applies to the second conductor **50**) is larger than the width in the X-axis direction of the second core **30**, and is equal to or smaller than the width in the X-axis direction of the first core **20**.

(35) As shown in FIG. **2**, the first conductor **40** includes a first main body portion **41** and the first mounting portions **42***a*, **42***b*. The first main body portion **41** has a substantially flat plate shape and is formed in a longitudinal shape in the X-axis direction. As shown in FIGS. **3**A and **3**B, the first main body portion **41** is arranged inside a space defined by the first recess portion **24***a* of the first core **20** and the second recess portion **34***a* of the second core **30**. More specifically, the first main body portion **41** extends in the X-axis direction inside the space without contacting the bottom surfaces of the first recess portion **24***a* and the second recess portion **34***a*. A gap is formed between the first main body portion **41** and each of the bottom surfaces of the first recess portion **24***a* and the second recess portion **34***a*.

(36) As shown in FIG. **2**, the first mounting portion **42***a* is formed on one end portion in the X-axis direction (a longitudinal direction) of the first main body portion **41**, and the first mounting portion **42***b* is formed on the other end portion in the X-axis direction (the longitudinal direction) of the first main body portion **41**. The first mounting portions **42***a*, **42***b* intersect the first main body portion **41** substantially perpendicularly and have surfaces parallel to a Y-Z plane.

(37) As shown in FIG. **3**C, the first mounting portion **42***a* is arranged along a side surface on one end side in the X-axis direction of the second core **30**, and the first mounting portion **42***b* is arranged along a side surface on the other end side in the X-axis direction of the second core **30**. A gap is formed between the mounting portions **42***a*, **42***b* and each side surface in the X-axis direction of the second core **30**. As shown in FIG. **1**B, lower ends of the mounting portions **42***a*, **42***b* are located below one end in the Z-axis direction of the second base portion **31**. Lands of a mounting substrate (not shown) are connected to the mounting portions **42***a*, **42***b* by joining members such as solder and conductive adhesive, and the coil device **10** can be connected to the mounting substrate via the mounting portions **42***a*, **42***b* (and the mounting portions **52***a*, **52***b*, which will be described later).

(38) As shown in FIG. **2**, the first mounting portion **42***a*, **42***b* respectively include first notch portions **420***a*, **420***b* and first lateral protruding portions **421***a*, **421***b*. The first notch portions **420***a*, **420***b* are formed on one end sides in the Y-axis direction of the first mounting portions **42***a*, **42***b* (a side on which the second conductor **50** is arranged). By the first notch portions **420***a*, **420***b*, a lower end portion located on the one end side in the Y-axis direction of each of the first mounting portion **42***a*, **42***b* is cut out at a predetermined depth toward an upper side in the Z-axis direction and the other end side in the Y-axis direction.

(39) The first lateral protruding portions **421***a*, **421***b* extend in the Y-axis direction toward a side on which the second outer leg portion **32***a* (that is, one of the pair of second outer leg portions **32***a*, **32***b*) is arranged. As shown in FIG. **3**C, a protruding width W**3** of the first lateral protruding portions **421***a*, **421***b* in the Y-axis direction is substantially equal to a width W**5** in the Y-axis direction of the first outer leg portions **22***a*, **22***b* and the second outer leg portions **32***a*, **32***b* shown in FIG. **1**B. However, W**3** may also be smaller than W**5**, and the protruding width W**3** may be appropriately determined within a range in which the first lateral protruding portions **421***a*, **421***b* do not protrude outside the second outer leg portion **32***a* in the Y-axis direction.

(40) As shown in FIG. **2**, the second conductor **50** includes a second main body portion **51** and the second mounting portions **52***a*, **52***b*. Since the second main body portion **51** has the same configuration as the first main body portion **41**, detailed description thereof will be omitted. The second main body **51** is arranged inside a space defined by the first recess portion **24***b* of the first core **20** and the second recess portion **34***b* of the second core **30**.

(41) The second mounting portions **52***a*, **52***b* are formed respectively on one end and the other end in the X-axis direction of the second conductor **50**, and respectively include second notch portions

520*a*, **520***b* and second lateral protruding portions **521***a*, **521***b*. The second notch portions **520***a*, **520***b* are formed on the other end sides in the Y-axis direction of the second mounting portions **52***a*, **52***b* (a side on which the first conductor **40** is arranged). By the second notch portions **520***a*, **520***b*, a lower end portion located on the other end side in the Y-axis direction of each of the second mounting portion **52***a*, **52***b* is cut out at a predetermined depth toward the upper side in the Z-axis direction and the one end side in the Y-axis direction.
(42) At positions where the first notch portions **420***a*, **420***b* and the second notch portions **520***a*, **520***b* are formed, it is possible to increase a distance between the first mounting portion **42***a* and the second mounting portion **52***a*. Therefore, when the coil device **10** is connected to the mounting substrate (not shown), a solder bridge is less likely to occur between the first mounting portion **42***a* and the second mounting portion **52***a*, and an accompanying short-circuit defect can also be prevented.
(43) The second lateral protruding portions **521***a*, **521***b* extend in the Y-axis direction toward a side on which the second outer leg portion **32***b* (that is, the other of the pair of second outer leg portions **32***a*, **32***b*) is arranged, and a protruding direction of the second lateral protruding portions **521***a*, **521***b* is opposite to a protruding direction of the first lateral protruding portions **421***a*, **421***b*. A protruding width of the second lateral protruding portions **521***a*, **521***b* in the Y-axis direction is substantially equal to the protruding width of the first lateral protruding portions **421***a* and **421***b* in the Y-axis direction.
(44) As shown in FIG. **1**B, according to the present embodiment, as magnetic body (magnetic materials), the first center leg portion **23** and the second center leg portion **33** are arranged between the first conductor **40** (the first mounting portions **42***a*, **42***b*) and the second conductor **50** (the second mounting portions **52***a*, **52***b*). In coupled inductors in the related art, no magnetic body is arranged between conductors from a viewpoint of enhancing magnetic coupling between conductors, whereas in the coil device **10** according to the present embodiment, between the first conductor **40** and the second conductor **50**, a magnetic body for reducing coupling therebetween is arranged (interposed).
(45) As described above, the first center leg portion **23** and the second center leg portion **33** are continuous respectively from one ends to the other ends in the X-axis direction of the first core **20** and the second core **30** (FIG. **3**C), and are joined with each other in the Z-axis direction. Therefore, the space in which the first main body portion **41** of the first conductor **40** is accommodated (the space surrounded by the first recess portion **24***a* and the second recess portion **34***a*) and the space in which the second main body portion **51** of the second conductor **50** is accommodated (the space surrounded by the first recess portion **24***b* and the second recess portion **34***b*) are substantially separated by the center leg portions **23** and **33** without communicating with each other (the above spaces only communicate with each other through a slight gap due to the first gap **61**).
(46) A ratio of a cross-sectional area of the center leg portions **23**, **33** along the Y-Z plane (a sum of cross-sectional areas of the first center leg portion **23** and the second center leg portion **33**) to a cross-sectional area of the outer leg portions **22***a*, **32***a* along the Y-Z plane (a sum of cross-sectional areas of the first outer leg portion **22***a* and the second outer leg portion **32***a*) is preferably 1:1 to 1:4. Similarly, a ratio of the cross-sectional area of the center leg portions **23**, **33** along the Y-Z plane (the sum of the cross-sectional areas of the first center leg portion **23** and the second center leg portion **33**) to a cross-sectional area of the outer leg portions **22***b*, **32***b* along the Y-Z plane (a sum of cross-sectional areas of the first outer leg portion **22***b* and the second outer leg portion **32***b*) is preferably 1:1 to 1:4.
(47) For example, it is possible to make a coupling coefficient between the first conductor **40** and the second conductor **50** to approximately 0.14 to 0.24 by setting the above ratios to approximately 1:1. It is possible to make the coupling coefficient between the first conductor **40** and the second conductor **50** to approximately 0.25 to 0.35 by setting the above ratios to approximately 1:2. It is possible to make the coupling coefficient between the first conductor **40** and the second conductor

50 to approximately 0.45 to 0.55 by setting the above ratios to approximately 1:4.

(48) Therefore, by adjusting the above ratios to any ratio between 1:1 and 1:4 (however, the cross-sectional area of the center leg portions **23**, **33** is smaller than the cross-sectional area of the outer leg portions **22***a*, **32***a* or **22***b*, **32***b*), the coupling coefficient between the first conductor **40** and the second conductor **50** can be adjusted to a desired value as described above. By setting the above ratios to approximately 2:1, it is possible to reduce the coupling coefficient between the first conductor **40** and the second conductor **50** to approximately 0.13 to 0.17, and if necessary, the cross-sectional area of the center leg portions **23**, **33** may be larger than the cross-sectional area of the outer leg portions **22***a*, **32***a* or **22***b*, **32***b*.

(49) In the present embodiment, the lengths L**1**, L**4** in the Z-axis direction of the outer leg portions **22***a*, **32***a*, the lengths L**2**, L**5** in the Z-axis direction of the outer leg portions **22***b*, **32***b*, and the lengths L**3**, L**6** in the Z-axis direction of the center leg portions **23**, **33** are substantially equal to each other. Therefore, by setting a ratio of the width W**4** of the center leg portions **23**, **33** to the width W**5** of the outer leg portions **22***a*, **32***a* (or the outer leg portions **22***b*, **32***b*) to 1:1 to 1:4, the ratio of the cross-sectional area of the center leg portions **23**, **33** to the cross-sectional area of the outer leg portions **22***a*, **32***a* (or the outer leg portions **22***b*, **32***b*) can be set to 1:1 to 1:4.

(50) The width W**4** of the center leg portions **23**, **33** is preferably 0.3 to 2.0 mm. The width W**5** of the outer leg portions **22***a*, **32***a* (or the outer leg portions **22***b*, **32***b*) is preferably 0.3 to 8.0 mm.

(51) In the present embodiment, since the magnetic body (center leg portions **23**, **33**) are arranged between the first conductor **40** and the second conductor **50**, magnetic fields generated from the first conductor **40** and the second conductor **50** pass through insides of the magnetic body (center leg portions **23**, **33**) arranged between the first conductor **40** and the second conductor **50**.

(52) In manufacture of the coil device **10**, the first core **20** and the second core **30** shown in FIG. **2** are prepared while the first conductor **40** and the second conductor **50** are prepared. Next, the first main body portions **41**, **51** of the conductors **40**, **50** are arranged inside the second recess portions **34***a*, **34***b* (or the first recess portions **24***a*, **24***b*) of the second core **30** (or the first core **20**). Then, by combining the first center leg portion **23** and the second center leg portion **33** and combining the first outer leg portions **22***a*, **22***b* and the second outer leg portions **32***a*, **32***b*, the first core **20** is combined with the second core **30**. At this time, the coil device **10** can be obtained by joining the first outer leg portions **22***a*, **22***b* and the second outer leg portions **32***a*, **32***b* with an adhesive or the like.

(53) In the coil device **10** according to the present embodiment, the magnetic body (center leg portions **23**, **33**) are arranged between the first conductor **40** and the second conductor **50**. In this case, the coupling between the first conductor **40** and the second conductor **50** is lower than that in a case where no magnetic body is arranged between the first conductor **40** and the second conductor **50**, and it is possible to reduce the magnetic coupling between the first conductor **40** and the second conductor **50**. By arranging the magnetic body between the first conductor **40** and the second conductor **50**, the magnetic body contribute to inductance of the coil device **10**, and the inductance value of the whole coil device **10** can be increased. Therefore, according to the coil device **10** in the present embodiment, it is possible to reduce the magnetic coupling between the first conductor **40** and the second conductor **50** while ensuring good inductance characteristics.

(54) In the present embodiment, the ratio of the cross-sectional area of the center leg portions **23**, **33** to the cross-sectional area of the outer leg portions **22***a*, **32***a* (or the outer leg portions **22***b*, **32***b*) is 1:1 to 1:4. In this case, the center leg portions **23**, **33** function as the above-mentioned magnetic body arranged between the first conductor **40** and the second conductor **50**. By such a configuration, it is possible to sufficiently reduce the coupling coefficient between the first conductor **40** and the second conductor **50**, and reduce the magnetic coupling between the first conductor **40** and the second conductor **50** while ensuring good inductance characteristics.

(55) In the present embodiment, the ratio of the width of the center leg portions **23**, **33** to the width of the outer leg portions **22***a*, **32***a* (or the outer leg portions **22***b*, **32***b*) is 1:1 to 1:4. When the ratio

of the cross-sectional area of the center leg portions **23**, **33** to the cross-sectional area of the outer leg portions **22***a*, **32***a* (or the outer leg portions **22***b*, **32***b*) is 1:1 to 1:4, by setting the ratio of the width of the center leg portions **23**, **33** to the width of the outer leg portions **22***a*, **32***a* (or the outer leg portions **22***b*, **32***b*) within the above-mentioned range, it is possible to match the protruding widths of the center leg portions **23**, **33** and the outer leg portions **22***a*, **32***a* (or the outer leg portions **22***b*, **32***b*) with each other, and a symmetry between the first core **20** and the second core **30** is good. Therefore, the coil device **10** having good inductance characteristics can be effectively obtained.

(56) In the present embodiment, the first core **20** is arranged above the second core **30** and is larger than the second core **30**. Therefore, when the first conductor **40** and the second conductor **50** are arranged between the first core **20** and the second core **30**, it is possible to prevent the first conductor **40** and the second conductor **50** from protruding outside the first core **20**, which can contribute to miniaturization of the coil device **10**.

(57) In the present embodiment, the first mounting portions **42***a*, **42***b* are provided at end portions in a longitudinal direction of the first conductor **40**, and the second mounting portions **52***a*, **52***b* are provided at end portions in a longitudinal direction of the second conductor **50**. The first mounting portions **42***a*, **42***b* extend toward a side on which the outer leg portions **22***a*, **32***a* are arranged, and the second mounting portions **52***a*, **52***b* extend toward a side on which the outer leg portions **22***b*, **32***b* are arranged, which is opposite to that of the first mounting portions **42***a*, **42***b*. Therefore, the first mounting portions **42***a*, **42***b* and the second mounting portions **52***a*, **52***b* can be separated from each other, and it is possible to prevent a short circuit failure from occurring between the first mounting portions **42***a*, **42***b* and the second mounting portions **52***a*, **52***b*. It is further possible to ensure a sufficient mounting area for each of the first mounting portions **42***a*, **42***b* and the second mounting portions **52***a*, **52***b*, and the coil device **10** can be firmly fixed to the mounting substrate (not shown).

(58) In the present embodiment, at least one of the first core **20** and the second core **30** contains the metallic magnetic material. Therefore, the coupling coefficient between the first conductor **40** and the second conductor **50** can be effectively reduced to a desired value (for example, preferably approximately 0.1 to 0.5, more preferably approximately 0.3 to 0.5).

(59) In the present embodiment, the first conductor **40** and the second conductor **50** are made of the conductive plate sheets. Therefore, permissible currents flowing through the first conductor **40** and the second conductor **50** can be increased.

Second Embodiment

(60) A coil device **110** according to a second embodiment shown in FIG. **4** has the same configuration as that of the coil device **10** according to the first embodiment except for the following points, and exhibits the same effects. In FIG. **4**, members common to members in the coil device **10** according to the first embodiment are denoted by common reference numerals, and a part of description thereof will be omitted.

(61) As shown in FIG. **4**, the coil device **110** includes a first core **120** and a second core **130**. The first core **120** has a substantially flat plate shape (substantially rectangular parallelepiped shape) and constitutes a so-called I-shaped core.

(62) The second core **130** includes second outer leg portions **132***a*, **132***b*, a second center leg portion **133**, and second recess portions **134***a*, **134***b*. A length in the Z-axis direction of the second outer leg portions **132***a*, **132***b* is larger than the length in the Z-axis direction of the second outer leg portions **32***a*, **32***b* according to the first embodiment. A length of the center leg portion **133** is larger than the length in the Z-axis direction of the center leg portion **33** according to the first embodiment. A depth in the Z-axis direction of the second recess portions **134***a*, **134***b* is larger than the depth in the Z-axis direction of the second recess portions **34***a*, **34***b* according to the first embodiment.

(63) In the present embodiment, at least one of the first core **120** and the second core **130** (only the

second core **130** according to an example shown in the drawing) has an E-shaped core having the second center leg portion **133** and a pair of the second outer leg portions **132***a*, **132***b*. The first core **120** may be configured to an E-shaped core, and the second core **130** may be configured to an I-shaped core.

(64) By such a configuration, it is possible to combine the I-shaped first core **120** and the E-shaped second core **130** with the gaps **61**, **62** therebetween, and the coil device **110** can have an EI shape. In the present embodiment, a magnetic body arranged between the first conductor **40** and the second conductor **50** is only the second center leg portion **133**.

(65) In the present embodiment, the magnetic body (second center leg portion **133**) is arranged between the first conductor **40** and the second conductor **50**. Therefore, the same effects as those of the first embodiment can be obtained.

Third Embodiment

(66) A coil device **210** according to a third embodiment shown in FIG. **5** has the same configuration as that of the coil device **10** according to the first embodiment except for the following points, and exhibits the same effects. In FIG. **5**, members common to members in the coil device **10** according to the first embodiment are denoted by common reference numerals, and a part of description thereof will be omitted.

(67) As shown in FIG. **5**, the coil device **210** includes a second core **230**. The second core **230** differs from the second core **30** according to the first embodiment in that the second core **230** includes a protruding portion **35**. The protruding portion **35** protrudes in an X-axis direction from a side surface on one end side in the X-axis direction of the second core **230** toward outside of the second core **230**. Although not shown, a protruding portion protruding in the X-axis direction toward the outside of the second core **230** is also formed on a side surface on the other end side in the X-axis direction of the second core **230**.

(68) The protruding portion **35** straddles the second base portion **31** and the second center leg portion **33** in a Z-axis direction. That is, by the protruding portion **35**, substantially central portions in a Y-axis direction of the second center leg portion **33** and the second base portion **31** protrude in the X-axis direction toward the outside of the second core **230**.

(69) A width of the protruding portion **35** in the Y-axis direction is substantially the same as the width of the first center leg portion **23** and the second center leg portion **33** in the Y-axis direction. A length of the protruding portion **35** in the Z-axis direction is substantially equal to a sum of the lengths of the second base portion **31** and the second center leg portion **33** in the Z-axis direction. A protruding width of the protruding portion **35** in the X-axis direction is substantially equal to the length L**7** shown in FIG. **1**C. At a position where the protruding portion **35** is formed, a side surface on one end side in the X-axis direction of the first core **20** and a side surface on one end side in the X-axis direction of the second core **230** are substantially flush with each other.

(70) In the present embodiment, the first mounting portion **42***a* is arranged on one side in the Y-axis direction with the protruding portion **35** in between, and the second mounting portion **52***a* is arranged on the other side in the Y-axis direction with the protruding portion **35** in between. In this way, by arranging (interposing) the protruding portion **35** between the first mounting portion **42***a* and the second mounting portion **52***a*, it is possible to effectively prevent a short circuit failure from occurring between the first mounting portion **42***a* and the second mounting portion **52***a*.

(71) The present invention is not limited to the above-mentioned embodiments and various modifications can be made within a scope of the present invention.

(72) In each of the above-mentioned embodiments, an application example of the coil device **10** according to the present invention to a coupled inductor is described, but the present invention may also be applied to other inductors or other coil devices.

(73) In the first embodiment, the first core **20** and the second core **30** may be integrally formed (one core). In this case, the first gap **61** shown in FIG. **1**B may be omitted, and the second gap **62** may be further omitted. The same applies to the second embodiment and the third embodiment.

(74) In the first embodiment, the magnetic body arranged between the first conductor **40** and the second conductor **50** is a part (center leg portions **23**, **33**) of the cores **20**, **30**, but the magnetic body may be formed separately from the cores **20**, **30**. The same applies to the second embodiment and the third embodiment. For example, in the first embodiment, each of the cores **20**, **30** may be made of a flat plate-shaped core, and the first conductor **40** and the second conductor **50**, which are arranged adjacent to each other, may be sandwiched between the cores **20**, **30** with a separately prepared magnetic body arranged between the first conductor **40** and the second conductor **50**. The magnetic body used in this case may be, for example, a magnetic body corresponding to a shape of the center leg portion **133** (see FIG. **4**) according to the second embodiment. In this case, the magnetic body may be made of a material different from those of the first core **20** and the second core **30**.
(75) In the first embodiment, a height of the first center leg portion **23** from the bottom surfaces of the first recess portions **24***a*, **24***b* may be different from a height of the second center leg portion **33** from the bottom surfaces of the second recess portions **34***a*, **34***b*. In this case, as shown in FIG. **1**B, when the first core **20** and the second core **30** are combined, the joint portion between the first center leg portion **23** and the second center leg portion **33** is arranged at any height position between the bottom surfaces of the first recess portions **24***a*, **24***b* and the bottom surfaces of the second recess portions **34***a*, **34***b*, so that the first center leg portion **23** and the second center leg portion **33** can be arranged between the first conductor **40** and the second conductor **50**. Therefore, in this case as well, it is possible to reduce the magnetic coupling between the first conductor **40** and the second conductor **50** while ensuring good inductance characteristics. The same applies to the third embodiment.
(76) When the first core **20** and the second core **30** are combined, by the magnetic body arranged between the first conductor **40** and the second conductor **50**, it is preferable that 50% or more of an area in the Z-axis direction between the bottom surfaces of the first recess portions **24***a*, **24***b* and the bottom surfaces of the second recess portions **34***a*, **34***b* is occupied, and it is more preferable that 60% or more of the same area is occupied.
(77) In the first embodiment, although the lengths of the first core **20** and the second core **30** in the Z-axis direction are substantially equal to each other, the lengths may be different. The length L**1** and the length L**4** shown in FIG. **1**B may be different. The length L**3** and the length L**6** may be different. The length L**2** and the length L**5** may be different. The same applies to the second embodiment and the third embodiment.
(78) In the first embodiment, as shown in FIG. **1**A, the overall shape of the coil device **10** when the first core **20** and the second core **30** are combined is a substantially flat (thin) rectangular parallelepiped shape. However, for example, the length of the second core **30** in the Z-axis direction may be larger than the length of the first core **20** in the Z-axis direction, and the overall shape may be a cube shape. In this case, it is possible to increase the coupling coefficient between the first conductor **40** and the second conductor **50**.
(79) In this case, by adjusting the ratio of the cross-sectional area of the center leg portions **23**, **33** (the sum of the cross-sectional areas of the first center leg portion **23** and the second center leg portion **33**) to the cross-sectional area of the outer leg portions **22***a*, **32***a* (the sum of the cross-sectional areas of the first outer leg portion **22***a* and the second outer leg portion **32***a*) to any value within the range of 1:1 to 1:4, the coupling coefficient between the first conductor **40** and the second conductor **50** can be adjusted between approximately 0.2 to 0.5. In this case, the materials constituting the first core **20** and the second core **30** may be changed as necessary.
(80) As shown in FIG. **1**A, when the coil device **10** has a thin shape, a sum of the length L**3** of the first center leg portion **23** in the Z-axis direction and the length L**6** of the second center leg portion **33** in the Z-axis direction shown in FIG. **1**B is preferably 0.55 to 0.75 mm. A length of the first base portion **21** or the second base portion **31** in the Z-axis direction is preferably 0.25 to 0.4 mm.
(81) In each of the above-mentioned embodiments, the first notch portions **420***a*, **420***b* and the

second notch portions **520**_a_, **520**_b_ may be omitted from the first mounting portions **42**_a_, **42**_b_ and the second mounting portions **52**_a_, **52**_b_, respectively.
(82) In each of the above-mentioned embodiments, the first conductor **40** and the second conductor **50** may be made of conductors (for example, wires) other than the conductive plate sheets.
REFERENCE SIGNS LIST
(83) **10**, **110**, **210** coil device **20**, **120** first core **21** first base portion **22**_a_, **22**_b_ first outer leg portion **23** first center leg portion **24**_a_, **24**_b_ first recess portion **30**, **130**, **230** second core **31** second base portion **32**_a_, **32**_b_, **132**_a_, **132**_b_ second outer leg portion **33**, **133** second center leg portion **34**_a_, **34**_b_, **134**_a_, **134**_b_ second recess portion **35** protruding portion **40** first conductor **41** first main body portion **42**_a_, **42**_b_ first mounting portion **420**_a_, **420**_b_ first notch portion **421**_a_, **421**_b_ first lateral protruding portion **50** second conductor **51** second main body portion **52**_a_, **52**_b_ second mounting portion **520**_a_, **520**_b_ second notch portion **521**_a_, **521**_b_ second lateral protruding portion **61** first gap **62** second gap

Claims

1. A coil device, comprising: a first core; a second core combined with the first core and having a main surface configured to face a mounting substrate and a side surface perpendicular to the main surface; a first conductor and a second conductor arranged adjacent to each other between the first core and the second core; and a magnetic body provided integrally or separately with the first core or the second core, wherein at least one of the first core and the second core includes (i) a center leg portion located between the first conductor and the second conductor and constituting the magnetic body and (ii) a pair of outer leg portions arranged on both sides of the center leg portion, the first conductor has (i) a first main body portion arranged between the first core and the second core and (ii) a first mounting portion continuous to the first main body portion and arranged over the side surface, the second conductor has (i) a second main body portion arranged between the first core and the second core and (ii) a second mounting portion continuous to the second main body portion and arranged over the side surface, the first mounting portion has a first lateral protruding portion protruding on one side in a first direction (Y-axis direction) from the first main body portion and the second mounting portion has a second lateral protruding portion protruding on the other side in the first direction (Y-axis direction) from the second main body portion, and the first mounting portion and the second mounting portion extend parallel to the side surface as a whole and lower ends of the first lateral protruding portion and the second lateral protruding portion protrude downward from the main surface in a direction from the first core to the second core along the side surface without bending.
2. The coil device according to claim 1, wherein an end portion in a second direction (X-axis) perpendicular to the first core in the first direction protrudes outward from an end portion of the center leg portion of the second core in the second direction.
3. The coil device according to claim 1, wherein a ratio of a cross-sectional area of the center leg portion to a cross-sectional area of the outer leg portions is 1:1 to 1:4.
4. The coil device according to claim 3, wherein a ratio of a width of the center leg portion to a width of the outer leg portions is 1:1 to 1:4.
5. The coil device according to claim 1, wherein the first core is arranged above the second core and is larger than the second core.
6. The coil device according to claim 3, wherein the first core is arranged above the second core and is larger than the second core.
7. The coil device according to claim 4, wherein the first core is arranged above the second core and is larger than the second core.
8. The coil device according to claim 1, wherein the first core includes a first center leg portion and first outer leg portions; the second core includes a second center leg portion and second outer leg

portions; a first recess portion is formed between the first center leg portion and each of the first outer leg portions; a second recess portion is formed between the second center leg portion and each of the second outer leg portions; and a height of the first center leg portion from a bottom surface of the first recess portion is different from a height of the second center leg portion from a bottom surface of the second recess portion.
9. The coil device according to claim 3, wherein the first core includes a first center leg portion and a first outer leg portion; the second core includes a second center leg portion and a second outer leg portion; a first recess portion is formed between the first center leg portion and the first outer leg portion; a second recess portion is formed between the second center leg portion and the second outer leg portion; and a height of the first center leg portion from a bottom surface of the first recess portion is different from a height of the second center leg portion from a bottom surface of the second recess portion.
10. The coil device according to claim 4, wherein the first core includes a first center leg portion and a first outer leg portion; the second core includes a second center leg portion and a second outer leg portion; a first recess portion is formed between the first center leg portion and the first outer leg portion; a second recess portion is formed between the second center leg portion and the second outer leg portion; and a height of the first center leg portion from a bottom surface of the first recess portion is different from a height of the second center leg portion from a bottom surface of the second recess portion.
11. The coil device according to claim 1, wherein the first mounting portion includes two parts provided at respective end portions in the second direction of the first conductor; and the second mounting portion includes two parts provided at respective end portions in the second direction of the second conductor.
12. The coil device according to claim 3, wherein the first mounting portion includes two parts provided at respective end portions in the second direction of the first conductor; and the second mounting portion includes two parts provided at respective end portions in the second direction of the second conductor.
13. The coil device according to claim 4, wherein the first mounting portion includes two parts provided at respective end portions in the second direction of the first conductor; and the second mounting portion includes two parts provided at respective end portions in the second direction of the second conductor.
14. The coil device according to claim 1, wherein at least one of the first core and the second core contains a metallic magnetic material.
15. The coil device according to claim 3, wherein at least one of the first core and the second core contains a metallic magnetic material.
16. The coil device according to claim 4, wherein at least one of the first core and the second core contains a metallic magnetic material.
17. The coil device according to claim 1, wherein the first conductor and the second conductor are made of conductive plate sheets.
18. The coil device according to claim 3, wherein the first conductor and the second conductor are made of conductive plate sheets.
19. The coil device according to claim 4, wherein the first conductor and the second conductor are made of conductive plate sheets.
20. A coil device, comprising: a first core; a second core combined with the first core and having a main surface configured to face a mounting substrate and a side surface perpendicular to the main surface; and a first conductor and a second conductor arranged adjacent to each other in a first direction between the first core and the second core, the side surface being at an end of the second core in a second direction perpendicular to the first direction, the second core being arranged below the first core in a third direction, the main surface being at a bottom of the second core facing away from the first core in the third direction, the third direction being perpendicular to the first direction

and the second direction, wherein at least one of the first core and the second core includes (i) a center leg portion located between the first conductor and the second conductor in the first direction and constituting a magnetic body and (ii) a pair of outer leg portions arranged on both sides of the center leg portion in the first direction, the first conductor has (i) a first main body portion arranged between the first core and the second core in the third direction and (ii) a first mounting portion continuous to the first main body portion and arranged over the side surface, the first mounting portion being at an end of the first main body portion in the second direction, the second conductor has (i) a second main body portion arranged between the first core and the second core in the third direction and (ii) a second mounting portion continuous to the second main body portion and arranged over the side surface, the second mounting portion being at an end of the second main body portion in the second direction, the first mounting portion and the second mounting portion being arranged adjacent to each other in the first direction over the side surface, the first mounting portion and the second mounting portion extending from between the first core and the second core toward the main surface in the third direction, and the first mounting portion has a first lateral protruding portion protruding on one side of the first direction from the first main body portion and the second mounting portion has a second lateral protruding portion protruding on the other side of the first direction from the second main body portion, and the first mounting portion and the second mounting portion extend parallel to the side surface as a whole and lower ends of the first lateral protruding portion and the second lateral protruding portion in the third direction protrude downward from the main surface in the third direction along the side surface without bending.